



Assignment: Individual PM

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GIK2PG

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Individual PM – Reflection on System Development Process

1. Introduction

This PM describes how the project was executed, with a primary focus on the development process rather than the final product itself. The project was carried out in collaboration with Trafikverket and involved developing a digital *Escape Room* application intended primarily for use at fairs. The purpose of the application was to engage students through interactive mini games and quizzes.

The aim of this document is to reflect on how the project was planned, managed and executed, as well as to analyze the tools, collaboration strategies and individual contributions involved. The discussion is structured according to the required questions and emphasizes lessons learned throughout the project.

2. Planning and Monitoring

a. How was the project planned?

The project was planned to use an agile and iterative approach. Rather than following a strictly predefined plan, we continuously adapted our workflow based on feedback and newly identified challenges.

At an early stage, the group established a working agreement to clarify expectations, responsibilities and communication practices. To improve efficiency, the team was divided into two subgroups, one responsible for the higher education version of the application and the other for the high school version in which I participated.

Planning was primarily conducted using Trello, where tasks were organized into a backlog and broken down into smaller, manageable work items. Decisions were made collectively during team meetings, meaning that the project was driven by consensus rather than strong centralized leadership.

b. How often did plans change and why?

The project plan was revised multiple times throughout the development process, which is consistent with agile methodologies.

The main causes for these changes were technical challenges, particularly during the initial setup of the backend environment and its integration with the frontend and database. As development progressed, unforeseen bugs and system inconsistencies required us to temporarily shift focus from feature development to system stabilization.

Additionally, design decisions were revisited based on user and stakeholder considerations. For example, one of the planned mini games was replaced entirely after it was deemed unsuitable for the intended target group.

These changes highlight the dynamic nature of the development process and the necessity of maintaining flexibility.

c. Was the planning accurate and useful?

While the planning was not always precise in terms of time estimation, it proved to be highly useful as a guiding structure.

The agile approach allowed us to manage uncertainty effectively and the use of smaller, clearly defined tasks ensured that progress could be maintained even when unexpected issues appeared.

Rather than serving as a rigid schedule, the planning functioned as a flexible framework that supported continuous development and adaptation.

d. Did the group manage changes well?

Overall, the group handled changes effectively.

We maintained open communication and demonstrated a strong collaborative mindset. When technical challenges appeared, team members actively supported each other through practices such as pair programming and joint debugging sessions.

This collaborative approach enabled us to respond to changes without significantly disrupting the overall progress of the project.

3. Source Code Management

a. Which system was used?

GitHub was used as the primary system for source code management.

A branching strategy was adopted, where new features were developed in separate branches before being merged into the main branch. This approach minimized the risk of introducing errors into the stable version of the application.

b. Problems encountered and solutions

One of the main challenges encountered was merge conflicts, which occurred when multiple developers modified the same files simultaneously.

These conflicts were primarily resolved through collaborative efforts, often involving screen sharing sessions via Zoom where the team could jointly review and resolve inconsistencies in the code.

Another significant challenge was ensuring that the development environment functioned consistently across all team members' machines. Given the integration of frontend and backend components, even minor configuration differences could lead to issues.

This was addressed by creating clear setup instructions and standardized scripts, ensuring that all team members could run the project in a consistent manner.

4. Other Tool Support

a. Which additional tools were used?

In addition to GitHub, several other tools were used to support the project:

- **Trello** for task management and backlog organization
- **Miro** for design, prototyping and visualization of system flows
- **Zoom** for communication, meetings and collaborative debugging

These tools collectively enabled efficient coordination and supported both technical and organizational aspects of the project.

b. Problems encountered and solutions

While the tools themselves functioned reliably, challenges appeared in maintaining consistent usage, particularly with Trello.

During periods of intensive development, updates to the task board were occasionally neglected. This issue was mitigated by actively incorporating Trello into team meetings, ensuring that the backlog remained up to date and aligned with the current state of the project.

5. Individual Contribution

a. Personal contributions

i. Artefacts produced

I contributed significantly to the development of the source code, particularly for the high school version of the application. This included implementing React components for the frontend and developing backend logic using C# and .NET.

ii. Architectural contributions

I was responsible for key components of the game engine, including the implementation of a global timer, a penalty system and a session-based anti-cheating mechanism.

Additionally, I contributed to the development of the admin dashboard, including dynamic configuration editors and a media management system.

iii. Planning and administration

I participated in translating technical requirements into actionable tasks within Trello and contributed to identifying dependencies between different system components.

iv. Roles

My primary role was that of a system developer, with a particular focus on frontend architecture and backend integration.

v. other contributions

I actively supported other team members by assisting with debugging, sharing technical insights and encouraging collaborative practices such as pair programming.

b. Reflection: strengths and areas for improvement

I consider my technical contributions to be one of my main strengths in this project. I took responsibility for complex components and successfully implemented solutions that were both robust and user-friendly.

However, there are also areas for improvement. One of the most significant challenges was communication across subgroups. Since the team was divided, there were moments of uncertainty regarding how different parts of the system would be integrated.

In future projects, I would advocate for more structured cross-team communication, such as short daily stand-up meetings involving all team members. This would improve alignment and reduce integration related challenges.

Furthermore, I would place greater emphasis on continuous documentation throughout the development process to enhance clarity and facilitate collaboration.

6. Overall Reflection

This project has provided valuable insight into how a real-world system development project is conducted.

In addition to strengthening my technical skills, I have developed a deeper understanding of the importance of planning, communication and adaptability. One key takeaway is that projects rarely proceed exactly as planned and the ability to respond effectively to change is essential.

Overall, the project has been both challenging and rewarding and it has contributed significantly to my development as a software engineer.

Appendix: Weekly Log

Week 4

The project commenced with a series of presentations where different organizations introduced their proposed project ideas. Following the submission of an interest form, our group was assigned a project in collaboration with Trafikverket.

We received an initial briefing outlining the objective of developing a digital *Escape Room* application. At this stage, I began reflecting on potential technical frameworks and architectures that would be suitable for implementing interactive and scalable game components.

Week 5

During this week, we conducted our initial kickoff meeting, where we discussed how to structure the project and distribute responsibilities within the group.

We established a formal working agreement to clarify expectations, communication practices and individual responsibilities. I took on the role of system developer and programmer, which became my primary focus throughout the project.

Week 6

The week began with a brainstorming session in which all team members proposed ideas for potential mini games. These concepts were presented and collectively evaluated, after which the most promising ideas were selected.

To improve efficiency, the team was divided into smaller subgroups. I joined the subgroup responsible for developing the high school version of the application. Additionally, we established a plan for recurring weekly meetings with the client to ensure continuous feedback and alignment.

Week 7

This week marked the beginning of the technical implementation phase. We set up the development environment, using React for the frontend and .NET for the backend and created a shared GitHub repository for version control.

Initial challenges arose during the setup of the backend environment and dependency management. These issues were resolved through careful examination of documentation and iterative debugging.

In parallel, we used Miro to design and visualize the game flow, which provided a clear conceptual foundation for further development.

Week 8

During this phase, I began implementing the core logic for several mini games. This included developing early prototypes for features such as puzzles and quiz elements.

A key focus was establishing a well-structured component architecture in React, ensuring that the codebase remained modular, maintainable and easily testable by other team members through GitHub.

Week 9

We held a progress review meeting with the instructors and the overall structure of the project began to take shape.

While some team members focused on documentation, such as the System Architecture Document (SAD), I dedicated most of my time to development. I refined the user interface, adjusted design elements and integrated Trafikverket's branding into the application.

Collaboration remained an important aspect of the workflow and I frequently participated in Zoom sessions where we shared screens and collaboratively debugged issues related to rendering and functionality.

Week 10

My primary focus during this week was the implementation of the game engine's timing functionality and its integration with the admin interface.

I developed a global timer hook (*useGameTimer*) that tracks the player's progress across the entire session, including a penalty system that adds time for incorrect answers.

Additionally, I implemented a session-based mechanism using *sessionStorage* to ensure correct timer initialization and to prevent cheating by restricting unauthorized navigation through direct URL access.

Week 11

The main objective this week was system integration. At this stage, some challenges emerged in communication between the two subgroups, which led to uncertainty regarding data flow and integration between different parts of the system.

I contributed to the admin panel by developing a “Smart Editor” for sorting-based mini games. This feature allows non-technical users to configure game content through simple text inputs, which are internally converted into structured and secure JSON data.

Based on client feedback, we also replaced one of the planned mini games with a redesigned traffic light-based game.

Week 12

During this week, I completed the implementation of an image gallery component (*ImageGallery.jsx*) within the admin panel. This feature supports uploading, listing, copying and deleting media files, with corresponding API endpoints implemented in the backend.

We also introduced informational modal dialogs between game stages to enhance the user experience.

From a version control perspective, I performed a major fast-forward merge to synchronize my development branch with the main branch. Additionally, I cleaned up outdated database migrations and finalized the remaining implementation details in preparation for the final presentation.

Week 13

The final week focused on preparing and delivering the project presentation. Responsibilities were distributed among team members and the final demonstration was conducted on-site with Trafikverket present.

The project received very positive feedback and the client expressed interest in testing the application in a real-world event setting. A follow-up meeting was scheduled to facilitate the handover of the system.

Reflecting on the project, it was highly rewarding to observe how the individual components we developed came together into a functional system.